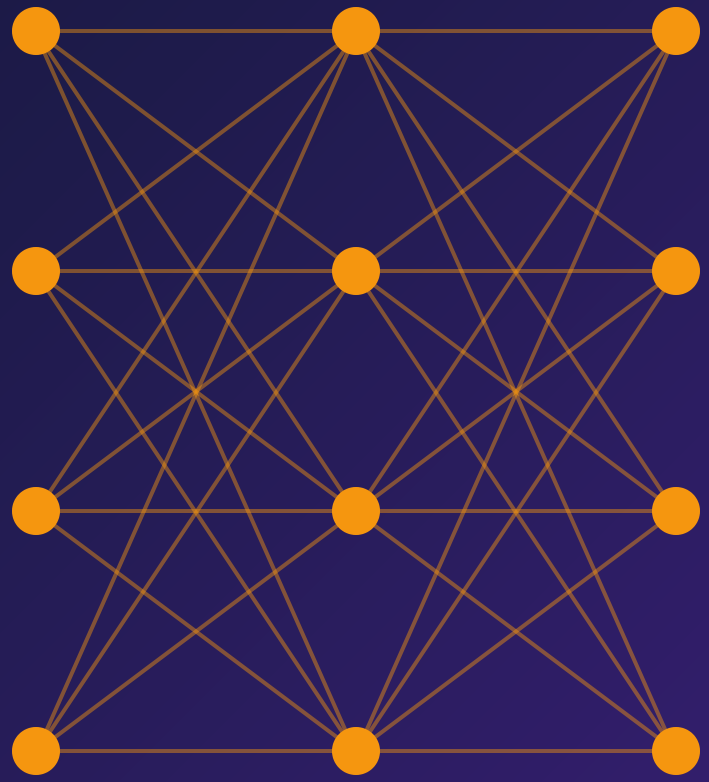
October 1 version

Open Model Evaluation



Four open-weight models, their architectures, and practical use cases

Team Name: Puranpoli

Hackathon presentation

MODEL 1 OF 5

gpt-oss-20b

OpenAI

MODEL 1 OF 5 · FACTS AND ARCHITECTURE

gpt-oss-20b

Assigned to Member 1

How it works: a mixture-of-experts (MoE) model holds 21B parameters in total, but for each token only about 3.6B are active, so it runs lighter than its total size suggests.

Developer	OpenAI
Release date	2025-08-05
Total parameters	21B
Active parameters	3.6B
Architecture	Mixture of Experts (MoE)
Context length	128K tokens
License	Apache 2.0 plus OpenAI usage policy
Language note	Mostly English training data; no complete list stated
Intended uses	Reasoning, coding, general knowledge, tool use, lower-latency or local use
Stated limitations	Text-only; mostly English; 128K context; Harmony format required; developers may need additional safety measures

MODEL 1 OF 5 · PROVISIONAL RATING

gpt-oss-20b

3.8/5

Overall rating

Advantage: Good reasoning with lower-latency and local-use potential.

Disadvantage: Text-only, mostly English, with weaker multilingual performance.

Reasoning	8/10
Factual accuracy	7.5/10
Instruction following	8/10
Coding	8/10
Indian languages	6.5/10
Total	**38/50**

Review evidence: good reasoning for a smaller model; published evaluations suggest weaker multilingual performance than English.

Provisional desk-research estimate — not a completed 25-prompt test.

MODEL 2 OF 5

gpt-oss-120b

OpenAI

MODEL 2 OF 5 · FACTS AND ARCHITECTURE

gpt-oss-120b

Assigned to Member 2

How it works: a mixture-of-experts (MoE) model holds 117B parameters in total, but for each token only about 5.1B are active.

Developer	OpenAI
Release date	2025-08-05
Total parameters	117B
Active parameters	5.1B
Architecture	Mixture of Experts (MoE)
Context length	128K tokens
License	Apache 2.0 plus OpenAI usage policy
Language note	Mostly English training data; no complete list stated
Intended uses	Reasoning, coding, general knowledge, tool use, larger deployments
Stated limitations	Text-only; mostly English; 128K context; Harmony format required; developers may need additional safety measures

MODEL 2 OF 5 · PROVISIONAL RATING

gpt-oss-120b

Advantage

Strong math and reasoning for larger deployments.

Disadvantage

Text-only, mostly English, and more demanding to deploy.

4.2/5

Overall rating

Reasoning	9/10
Factual accuracy	8.5/10
Instruction following	8.5/10
Coding	8.5/10
Indian languages	7.5/10
Total	**42/50**

Review evidence: particularly strong math and reasoning in OpenAI evaluations; independent work reports multilingual weakness; community experiences vary by task.

Provisional desk-research estimate — not a completed 25-prompt test.

MODEL 3 OF 5

Qwen3.8-27B

Alibaba Qwen team

MODEL 3 OF 5 · FACTS AND ARCHITECTURE

Qwen3.8-27B

Assigned to Member 3

Developer	Alibaba Qwen team
Release date	2026-08-19
Total parameters	27B
Active parameters	N/A (dense)
Architecture	Dense causal LM with vision encoder
Context length	262K native; extendable to 1M with configuration
License	Apache 2.0
Language note	Exact supported-language list not stated
Intended uses	Coding, research, professional and agent tasks; text, image, and video understanding
Stated limitations	Extension requires configuration and may affect performance; language list not enumerated

How it works: a dense model uses all 27B parameters for every token. It is a causal language model with a vision encoder, so it also reads images and video.

MODEL 3 OF 5 · PROVISIONAL RATING

Qwen3.8-27B

Advantage

Strong coding and reasoning with vision capability and long context.

Disadvantage

Language coverage and extended-context performance are less certain.

4.25/5

Overall rating

Reasoning	9/10
Factual accuracy	8.5/10
Instruction following	8.5/10
Coding	9/10
Indian languages	7.5/10
Total	**42.5/50**

Review evidence: strong published coding and reasoning results for its size; results and speed depend on serving setup; language coverage is less certain.

Provisional desk-research estimate — not a completed 25-prompt test.

MODEL 4 OF 5

Gemma 4 31B-it

Google DeepMind

MODEL 4 OF 5 · FACTS AND ARCHITECTURE

Gemma 4 31B-it

Assigned to Member 4

How it works: a dense model uses all of its roughly 31B parameters for every token, with no expert routing.

Developer	Google DeepMind
Release date	2026-03-31
Total parameters	30.7B (marketed as 31B)
Active parameters	N/A (dense)
Architecture	Dense
Context length	256K tokens
License	Apache 2.0
Language note	Trained on 140+ languages; out-of-box support stated for 35+
Intended uses	Reasoning, coding, tool use, text and image tasks
Stated limitations	May give incorrect or outdated facts; can struggle with complex or ambiguous tasks; possible training-data bias

MODEL 4 OF 5 · PROVISIONAL RATING

Gemma 4 31B-it

4.15/5

Overall rating

Advantage: Balanced dense model with strong reasoning, coding, multilingual, and multimodal potential.

Disadvantage: Factual errors, ambiguity problems, and training-data bias remain possible.

Reasoning	8.5/10
Factual accuracy	8/10
Instruction following	8.5/10
Coding	8.5/10
Indian languages	8/10
Total	**41.5/50**

Review evidence: strong published math, coding, and reasoning benchmarks; factual errors and bias remain possible.

Provisional desk-research estimate — not a completed 25-prompt test.

MODEL 5 OF 5

Gemma 4 26B-A4B-it

Google DeepMind

MODEL 5 OF 5 · FACTS AND ARCHITECTURE

Gemma 4 26B-A4B-it

Assigned to Member 5

How it works: a mixture-of-experts (MoE) model holds 25.2B parameters in total, but for each token only about 3.8B are active. All weights still need to fit in memory.

Developer	Google DeepMind
Release date	2026-03-31
Total parameters	25.2B (marketed as 26B)
Active parameters	3.8B
Architecture	Mixture of Experts (MoE)
Context length	256K tokens
License	Apache 2.0
Language note	Trained on 140+ languages; out-of-box support stated for 35+
Intended uses	Reasoning, coding, tool use, text and image tasks
Stated limitations	Same general Gemma limits; total parameters still require adequate model memory despite 3.8B active

MODEL 5 OF 5 · PROVISIONAL RATING

Gemma 4 26B-A4B-it

Advantage

Efficient MoE model with multilingual and multimodal capability.

Disadvantage

Total model memory is still significant and some positive reports are anecdotal.

4/5

Overall rating

Reasoning	8/10
Factual accuracy	8/10
Instruction following	8/10
Coding	8/10
Indian languages	8/10
Total	**40/50**

Review evidence: efficient MoE design; positive language-learning and science reports are anecdotal.

Provisional desk-research estimate — not a completed 25-prompt test.